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Definitions

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Roof System Overview & Purpose

A roof system functions as a **water-shedding assembly**. The deck (typically plywood or OSB) supports the load; the **underlayment** forms a secondary barrier; flashing, drip edges and fasteners shed water at vulnerable joints; and the roof covering (shingles, tiles, metal panels or membranes) provides the primary weather barrier. Underlayment and flashing are not meant to be exposed long-term—roof coverings must be installed over them within the manufacturer’s exposure limit (often 30–90 days). A properly designed system also ventilates the attic or roof cavity to equalize temperature and moisture; the International Residential Code (IRC) requires 1 sq ft of net-free vent area per 300 sq ft of attic floor (split roughly 60 % intake at soffits and 40 % exhaust at ridge) ¹. Venting prevents ice dams in cold climates and reduces overheating in hot climates.

Types of Roofing Materials

Asphalt shingles

Asphalt-shingle roofs are the most common residential covering because they are inexpensive and fairly easy to install. The IRC requires asphalt shingles on slopes $\geq 2:12$ ². Slopes between **2:12 and 4:12** are considered *low-slope*; they demand a **double layer of underlayment** or a self-adhered membrane because shallow pitch slows drainage and increases the chance of wind-driven water intrusion ³. Standard “3-tab” shingles have a typical life of 15–20 years; thicker **architectural** or **designer** shingles can last 25–30 years or more and may carry manufacturer warranties up to 50 years. Shingles are combustible; local codes may require Class A fire-rated products.

Metal roofing

Metal roofs can be panels with exposed fasteners, standing-seam systems, or metal shingles. The IRC differentiates between **metal shingles**, requiring $\geq 3:12$ slope ⁴, and **standing-seam panels**, which may be installed at very low slopes when seams are sealed—down to **0.25:12** for continuous standing-seam systems ⁵. Metal is lightweight, non-combustible and long-lasting; painted steel or aluminum panels typically last 40–50 years, and premium copper or zinc systems may last 50–100 years. Panels must be fastened or clipped according to the manufacturer’s wind-zone tables; high winds may require closer

fastener spacing and secure anchorage. Condensation under metal panels can corrode the deck, so use appropriate underlayment and ventilation.

Clay or concrete tiles

Tiles are heavy and require a reinforced deck. According to the IRC, clay and concrete tile roofs require **≥2.5 in 12** slope; slopes **2.5–4 in 12** must use **two layers of underlayment**. Underlayment attachment and tile fastening must meet the design wind speed; perimeter tiles need at least one corrosion-resistant fastener per tile and additional fasteners where design wind speeds exceed 130 mph. Tile roofs are very durable—service lives of 50 years or more—and non-combustible. Tiles are brittle and may break under impact; choose Class 4 impact-rated tiles in hail-prone regions. Because tiles are porous, a well-sealed underlayment is critical for moisture protection.

Membrane (single-ply) roofing

Low-slope or flat roofs (slopes **<2:12**) require continuous waterproof membranes rather than water-shedding shingles. **Thermoset EPDM** (rubber), **thermoplastic TPO** and **PVC** membranes are common. The IRC permits these membranes on slopes **≥0.25:12** ⁶. EPDM is black (absorbs heat) and typically fully adhered with seam tapes; warranties are often 20–30 years. TPO and PVC are white reflective membranes welded with hot air; TPO is economical and widely used, while PVC offers superior chemical and grease resistance but at a higher cost. Ballasted or mechanically attached assemblies must follow the manufacturer's wind uplift design. Membranes must be protected with walkway pads at high-traffic areas and require regular inspection for seam degradation and punctures.

Underlayment Types

Underlayment provides a **secondary weather barrier**; it is not a permanent roof covering. Key types:

Type	Description	Typical Uses / Specs
Asphalt-saturated felt	Traditional paper felt impregnated with asphalt; sold in #15 or #30 weights (approx. 15 or 30 lb/100 sq ft). It is vapour-permeable (approx. 5 perms) and offers limited water resistance. For roof slopes 2:12–4:12 double coverage is required with 19 in. horizontal overlaps ⁷ . On steep roofs (≥4:12), a 2-in. side lap and 4-in. end lap are typical ⁸ . Fasten with staples or nails; plastic cap fasteners are recommended in high-wind areas ⁹ .	Economical for most steep-slope applications; not UV resistant and should be covered quickly.

Type	Description	Typical Uses / Specs
Synthetic underlayment	Polypropylene or polyester sheet with a spun-bonded surface. Products such as GAF Tiger Paw® or FeltBuster® weigh roughly 2–3 lb/100 sq ft ¹⁰. They are strong, lightweight and often have anti-skid surfaces ¹¹. Typical installation requires 3 in. side laps and 6 in. end laps ¹². Fasten with plastic-cap nails or staples at 12 in. on center along laps and 24 in. in the field; in high-wind zones spacing is reduced (6 in./12 in.) ¹³. Maximum exposure time is 30–90 days depending on product ¹¹.	Use on most sloped roofs; good tear resistance and walkability. UV exposure limits must be followed.
Self-adhered ice & water shield	Rubberized asphalt membrane with peel-off release film. It self-seals around fasteners and provides a watertight barrier. Install directly on the deck, starting at the eaves and extending at least 24 in. inside the exterior wall line ¹⁴. Overlap courses by 3 in.; in valleys, keep the membrane full width and extend 18 in. each side ¹⁵. It may not require fasteners; on steep slopes, temporary nails every 18 in. can help hold it until adhesion occurs ¹⁴.	Required by code in ice-dam regions and recommended at eaves, valleys, skylights, and roof-wall intersections. Exposure time is limited (often 30 days).

Flashing, Drip Edge, Valleys, Venting, Overlaps

Flashing. Step flashing at roof-to-wall intersections should extend **≥4 in. up the wall and 4 in. onto the roof** ¹⁶. In high-wind zones, codes may require continuous base flashing with **L-shaped flashing** sized 4 × 4 in. or greater ¹⁷. Kick-out flashing or diverters at the end of roof-to-wall intersections direct water into gutters and prevent rot.

Drip edge. ARMA recommends corrosion-resistant drip-edge metal extending **≥2 in. onto the roof deck** and projecting **≥¼ in. below** ¹⁸. At eaves, the drip edge goes **under** the underlayment; at rakes, it goes **over** ¹⁹. Fasten with roofing nails spaced **8–10 in. on center**.

Valleys. For metal roof systems, install a **36 in. wide underlayment** centered in the valley so that **18 in.** covers each side, and lap successive pieces **6 in.** ²⁰. Ice-and-water shields should run full width down the valley and lap adjacent courses per manufacturer instructions ¹⁵. In tile roofs, valley flashings must be integrated with underlayment and provide weep openings to allow drainage.

Ventilation. Provide balanced intake and exhaust ventilation to comply with the **1 sq ft per 300 sq ft** rule ¹. For attic roofs, place vents in the soffit and at ridges; for low-slope membrane roofs, install scuppers or roof drains sized for local rainfall.

Overlaps and fastening. Always follow manufacturer instructions for lap widths and fastener patterns. For synthetic underlayment, maintain **3 in. side laps and 6 in. end laps** ¹² and stagger end laps by at least

3 ft. For felt on low-slope roofs, provide **19 in. overlap** to achieve double coverage ⁷ . Overlaps should run with the direction of water flow and never direct water under the lap ¹⁰ .

Installation Rules & Best Practices

- 1. Deck preparation.** Ensure the deck is clean, dry and properly fastened. Remove old nails or fasteners and repair rotten wood. A sealed deck (taped seams or full self-adhered membrane) improves wind- and water-resistance in coastal and high-wind zones.
- 2. Underlayment selection (Chain of Thought – reasoning example).**
Assess climate: In ice-dam regions, start with a self-adhered ice & water shield along eaves and valleys to seal around nails and resist water backups ¹⁴ . In warm, dry climates, synthetic underlayment may suffice.
Consider wind zone: High-wind areas require sealed roof decks and closer fastener spacing; choose underlayments with enhanced tear strength (synthetic or self-adhered).
Account for roof slope: Low slopes (2:12–4:12) require double coverage or self-adhered underlayment ³ , while slopes below 2:12 need a continuous membrane (TPO/EPDM/PVC).
Match roof material: Some metal roofing systems require high-temperature underlayment to withstand heat; tile roofs need two layers of felt or synthetic plus self-adhered membranes at penetrations.
- 3. Fastening.** Use plastic-cap nails or staples; avoid plain staples because they tear the underlayment. Follow the manufacturer’s nail spacing—common patterns are **12 in. on center** along laps and **24 in. in the field** for synthetic underlayment ¹² ; reduce to **6 in./12 in.** in high-wind zones ¹³ .
- 4. Weather exposure.** Underlayments are temporary barriers. Synthetic products may remain exposed for 30–90 days ¹¹ ; felt should be covered as soon as practical. Self-adhered ice & water shields should be covered within about 30 days ¹⁴ .
- 5. Penetrations and transitions.** At pipes, skylights, chimneys and roof-wall intersections, integrate underlayment with flashing. For self-adhered membranes, prime the deck if required and roll overlaps firmly to activate the adhesive ¹⁴ . In step-flashing applications, alternate shingle courses and metal step flashing; include a kick-out flashing at the base to divert water ¹⁶ .
- 6. Drip edge installation.** Apply drip edge metal **under** the underlayment at eaves and **over** at rakes ¹⁹ . Fasten at **8–10 in. on center** ¹⁸ .
- 7. Valley treatment.** Use ice-and-water shield or a metal valley liner; maintain adequate laps (6 in. or more) ²⁰ . For closed-cut shingle valleys, install an extra underlayment layer; for open metal valleys, leave a 2–4 in. open area for water flow.

Durability, Warranty & Material Selection Trade-offs

Material	Typical lifespan & warranty	Pros	Cons
3-tab asphalt shingles	15–20 yrs; basic 20–25 yr limited warranty	Low cost; easy to install; wide colour selection	Short lifespan; prone to wind damage; moderate fire rating

Material	Typical lifespan & warranty	Pros	Cons
Architectural or designer shingles	25–30 yrs+; 30–50 yr enhanced warranties	Thicker, dimensional appearance; better wind & impact resistance; can mimic slate/wood	Higher cost than 3-tab; still combustible
Standing-seam metal panels	40–50 yrs (steel/aluminium); up to 100 yrs (copper/zinc); 20–40 yr coatings	Long-lasting; non-combustible; sheds snow easily; low weight	Higher initial cost; expansion/contraction requires clips and slip joints; noise in rain
Clay/concrete tiles	50+ yrs; manufacturers often offer 50-yr limited warranties	Very durable; fireproof; high aesthetic value	Heavy (requires structural support); brittle under impact; expensive
EPDM membrane	20–30 yrs; 20–30 yr warranties	Proven track record; flexible in cold weather; fewer seams with large sheets	Dark surface absorbs heat; seams rely on tapes/adhesives; puncture risk
TPO membrane	15–25 yrs; 20–30 yr warranties	White reflective surface reduces cooling load; heat-welded seams; cost-effective	Newer product—long-term performance still maturing; requires skilled installers
PVC membrane	20–30 yrs; up to 30-yr warranties	Excellent chemical resistance; heat-welded seams; strong puncture resistance	Higher material cost; can shrink over time; plasticizer migration concerns

Likely User Questions

- **Do I need a synthetic underlayment?** Synthetic underlayment offers high tear resistance and longer exposure time than felt. For steep-slope roofs, either #30 felt or synthetic is acceptable; synthetics are recommended in high-wind or high-traffic situations. In low-slope (2:12–4:12) applications, double coverage or self-adhered membranes are required ³.
- **What is ice & water shield, and where is it required?** It is a self-adhered rubberized membrane that seals around fasteners and provides waterproof protection at eaves, valleys and penetrations. It should extend at least 24 in. inside the exterior wall line and overlap by 3 in. between courses ¹⁵. Many cold-climate codes require it on the entire eave to prevent ice-dam back-ups.
- **How far should underlayment overlap?** Felt underlayment overlaps 19 in. for low slopes and 2 in. for steep slopes ⁷. Synthetic underlayment typically requires 3 in. side laps and 6 in. end laps ¹². Ice-and-water shield courses overlap 3 in. and end laps at least 6 in. ¹⁵.

- **Does roof slope affect material choice?** Yes. Asphalt shingles require $\geq 2:12$ slope ²; metal shingles $\geq 3:12$; tiles $\geq 2.5:12$; standing-seam metal panels with sealed seams can be used down to 0.25:12 ⁵; single-ply membranes are required on slopes below 2:12 ⁶.
- **Can I eliminate felt underlayment entirely?** Some building codes allow synthetic underlayments or self-adhered membranes as substitutes for felt. However, removing underlayment entirely is risky; it provides temporary waterproofing and protects the deck from resin leaching and uneven surfaces. In very dry climates, some metal roof manufacturers may allow a “breathable” barrier or purlin systems, but this should be verified with the manufacturer.

Related Topics / Crosslinks

- **Roof Ventilation & Attic Insulation** – Balanced intake/exhaust venting reduces moisture and heat build-up ¹.
- **Step & Kickout Flashing** – Proper flashing prevents wall rot; step flashing should extend at least 4 × 4 in. and include kick-out diverters ¹⁶.
- **Drip Edge Installation** – Drip edges protect the deck edge and must be installed under/over underlayment correctly ¹⁹.
- **Tile Roofs in High-Wind Zones** – Use self-adhered membranes or two-layer underlayment and follow wind-load fastening tables.
- **Sealed Roof Decks & Hurricane Fortification** – The IBHS Fortified Roof™ program details sealed deck options and enhanced attachment for wind-resistant roofs.
- **Single-Ply Membrane Systems** – Understand differences between EPDM, TPO and PVC membranes, including seam technologies and insulation requirements.

Creative Explorations Using SCAMPER

- **Substitute:** Consider **breathable synthetic underlayments** in hot, humid climates where drying potential is important. For low-slope roofs, substitute traditional felt with **peel-and-stick membranes** to improve waterproofing.
- **Combine:** Combine **ice-and-water shield** with **synthetic underlayment** for a belt-and-suspenders approach on low-slope roofs—self-adhered at eaves/valleys and synthetic on the field.
- **Adapt:** Borrow from smart-home technology: integrate **moisture sensors** beneath underlayments to alert homeowners to leaks before they become visible. Adapt high-temperature automotive materials into roof membranes to improve heat resistance under metal roofs.
- **Modify:** Modify fastening patterns in high-wind zones—use closer nail spacing, ring-shank nails and sealed deck tapes. Modify overlaps by adding **butyl seam tape** at lap joints for additional water tightness on low-slope roofs ¹³.
- **Put to other use:** Use synthetic underlayment as a temporary **jobsite tarp** to protect decks or materials during construction thanks to its durability and UV resistance.
- **Eliminate:** In extremely arid climates with vented metal roofs, consider **eliminating traditional felt** and instead using batten systems that create an air space under the metal panels; verify with the manufacturer and code official.
- **Reverse:** Think about **reverse-lap** experiments in controlled mockups to see how water intrusion occurs when laps are incorrectly oriented; use the findings to train crews on proper lap direction.

1 PA_Crash_Course_Roof_Venting_FHB.pdf

https://buildingscience.com/sites/default/files/migrate/pdf/PA_Crash_Course_Roof_Venting_FHB.pdf

2 4 5 6 Measuring Roof Slope and Pitch - InterNACHI®

<https://www.nachi.org/roof-slope-pitch.htm>

3 Minimum Slope for Shingles What Contractors Need to Know

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7 9 Roofing Underlayment Types - InterNACHI®

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8 Mastering Roof Inspections: Underlayment, Part 2 - InterNACHI®

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16 17 Inspecting Step and Kickout Flashing at Roof-Wall Intersections - InterNACHI®

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